

LOOPS & FUNCTIONS

PART A - LOOPS

Definition

"loops execute code repeatedly until a condition becomes false."

Purpose

Loops reduce code repetition and improve automation

Types of Loops

- for loop
- while loop
- do-while loop
- for Each loop
- for...of loop

1. for Loop Syntax

```
for ( initialization ; condition ; increment ) {  
    code  
}
```

Example

```
for (let i = 1; i <= 5; i++) {  
  console.log(i);  
}
```

2. while Loop

```
let i = 1;  
while (i <= 5) {  
  console.log(i);  
  i++;  
}
```

3. do-while Loop

```
let i = 1;  
do {  
  console.log(i);  
  i++;  
} while (i <= 5);
```

4. for Each Loop

```
let fruits = ["Apple", "Mango"];  
fruits.forEach((fruit) => {  
  console.log(fruit);  
});
```

5. Break and Continue

1. Break

Stops the loop completely.

break;

2. Continue

Skips current iteration.

continue;

Real World Use

Loops are used in:

- displaying products
- notifications
- generating tables
- rendering posts
- dashboard data

PART B - FUNCTIONS

Definition

"Functions are reusable blocks of code that perform specific tasks."

Purpose

Functions improve:

- code reusability
- maintainability
- readability

1. Function Declaration

```
function greet() {  
    console.log("Hello");  
}
```

2. Function with Parameters

```
function add(a, b) {  
    return a + b;  
}
```

3. Arrow Function

```
const multiply = (a, b) => a * b;
```

4. Callback Function

```
function Process(callback) {  
    callback();  
}
```

5. Higher Order Function

"A function that accepts another function."

```
function execute(fn) {  
    fn();  
}
```

Real World Use

Functions are used in:

- authentication
- API calls
- payment systems
- forms
- Calculators

Important Points

- return sends data back.
- arrow functions are modern ES6 syntax.
- functions reduce duplicate code
- callbacks are important in async programming